

A Rigorous Single-Evaluation Test-Set Assessment of a SmallResNet Optimization Variant on PathMNIST

path-scientist-agent

1 Abstract

We report a frozen-candidate evaluation of a `SmallResNet` model (variant: `optimization`) on the PathMNIST dataset (SHA-256: `1e7fc200dd5aac79f39f0da26178be801ffac8e6d59c8533e51acebae953745a`). Training and tuning used only train/validation splits. After candidate freezing, the test split was evaluated exactly once; test results were not used for tuning, model selection, or retraining. The frozen candidate used learning rate 0.001, weight decay `1e-05`, OneCycle scheduling, label smoothing 0.1, and neither augmentation nor multiscale processing, across seeds 7, 17, and 27. The best validation Macro-F1 during tuning was 0.997589. On the single one-time test evaluation, the Macro-F1 mean was 0.834035 with standard deviation 0.075947, yielding a validation-to-test Macro-F1 gap of 0.158425. Seed 27 was unstable on test (Macro-F1 0.746367, best epoch 18), whereas seeds 7 and 17 reached Macro-F1 values of 0.875924 and 0.879813, respectively, both at best epoch 39. The three lowest-recall classes in the aggregate confusion matrix were class 7 (0.358), class 0 (0.724), and class 6 (0.858). We make no clinical use or diagnostic claims.

2 Introduction

This document reports a strictly controlled evaluation of a frozen model candidate on the PathMNIST dataset. The experimental protocol was designed to prevent any information leakage from the test split into model development. Specifically, training and tuning used only train/validation splits, and the frozen candidate was evaluated on the test split exactly once. Test results were not used for tuning, model selection, or retraining. The frozen candidate is a `SmallResNet` model with the `optimization` variant. The candidate was approved and frozen on 2026-08-18 by the project owner, with status `approved_frozen`. The dataset SHA-256 is `1e7fc200dd5aac79f39f0da26178be801ffac8e6d59c8533e51acebae953745a`. The primary metric is `macro_f1`. The test policy permits exactly one evaluation on the test split, with tuning feedback disallowed. The contributions of this report are: (i) a transparent record of the tuning grid and train/validation results; (ii) a single, post-freeze test evaluation; and (iii) a discussion of the observed validation-to-test gap and per-seed instability. We do not propose clinical use or diagnostic claims.

3 Related Work

This report is constrained to the supplied frozen artifacts and does not cite external literature. Within the supplied artifacts, the relevant prior context is the set of pre-freeze experiments on PathMNIST under the `pathmnist-m4` run group. These include the `main` phase variants (`baseline`, `augmentation`, `optimization`, `multiscale`, `combined`) and the `ablations` phase variants (`combined_no_augmentation`, `combined_no_multiscale`). The `optimization` variant was selected as the frozen candidate based on validation Macro-F1.

4 Materials and Methods

4.1 Dataset

The dataset is PathMNIST, identified by SHA-256 `1e7fc200dd5aac79f39f0da26178be801ffac8e6d59c8533e51acebae95374`. The dataset was partitioned into train, validation, and test splits. Training and tuning used only the train and validation splits. The test split was held out until after candidate freezing.

4.2 Model and Variant

The frozen candidate is a `SmallResNet` model with the `optimization` variant. The candidate status is `approved_frozen`, approved by the project owner on 2026-08-18.

4.3 Hyperparameters

The frozen hyperparameters are:

- Learning rate: `0.001`
- Weight decay: `1e-05`
- OneCycle: `true`
- Label smoothing: `0.1`
- Augmentation: `false`
- Multiscale: `false`

4.4 Seeds

Three seeds were used: 7, 17, and 27.

4.5 Primary Metric

The primary metric is `macro_f1`.

4.6 Test Policy

The test policy specifies:

- Split: `test`
- Evaluations allowed: `1`
- Tuning feedback allowed: `false`

The frozen candidate was evaluated on the test split exactly once after candidate freezing. Test results were not used for tuning, model selection, or retraining.

4.7 Runtime Environment

- PyTorch: `2.13.0+cu132`
- CUDA: `13.2`
- GPU: `NVIDIA GeForce RTX 5070 Ti Laptop GPU`

5 Experiments

5.1 Tuning

Six pre-freeze grid settings were completed. The best learning rate identified was 0.001, and the best weight decay was $1\text{e-}05$. The best validation Macro-F1 achieved during tuning was 0.997589.

5.2 Train/Validation Results

The train/validation results across phases and variants are summarized below. Each variant was run with three runs.

Phase	Variant	Runs	Macro-F1 mean	Macro-F1 std
main	baseline	3	0.956763	0.027502
main	augmentation	3	0.971311	0.011846
main	optimization	3	0.992459	0.008029
main	multiscale	3	0.978622	0.003913
main	combined	3	0.968218	0.024861
ablations	combined_no_augmentation	3	0.930088	0.036204
ablations	combined_no_multiscale	3	0.949365	0.020949

Among the **main** variants, **optimization** achieved the highest Macro-F1 mean (0.992459) with the lowest standard deviation (0.008029) among the non-multiscale variants. The **multiscale** variant had the lowest standard deviation (0.003913) but a lower mean (0.978622) than **optimization**.

5.3 Final One-Time Test Evaluation

After candidate freezing, the test split was evaluated exactly once. The evaluation count is 1. The test Macro-F1 mean was 0.834035, with a standard deviation of 0.075947. The validation-to-test Macro-F1 gap was 0.158425. Per-seed test results are:

Seed	Best epoch	Accuracy	Macro-F1	Loss
7	39	0.906128	0.875924	0.397548
17	39	0.906267	0.879813	0.412215
27	18	0.779666	0.746367	0.722006

6 Results

6.1 Aggregate Test Performance

The aggregate test Macro-F1 mean was 0.834035 with standard deviation 0.075947. The evaluation count was 1. The validation-to-test Macro-F1 gap was 0.158425.

6.2 Per-Seed Test Performance

Seeds 7 and 17 both achieved best epoch 39, with accuracies of 0.906128 and 0.906267, Macro-F1 values of 0.875924 and 0.879813, and losses of 0.397548 and 0.412215, respectively. Seed 27 achieved best epoch 18, with accuracy 0.779666, Macro-F1 0.746367, and loss 0.722006.

6.3 Aggregate Confusion Matrix

The aggregate confusion matrix over the three seeds is:

True \ Pred	0	1	2	3	4	5	6	7	8
0	2905	335	1	34	18	683	29	0	9
1	12	2529	0	0	0	0	0	0	0
2	0	2	958	0	8	28	0	2	19
3	0	0	3	1898	0	0	0	0	1
4	50	2	3	0	2873	76	44	16	41
5	2	120	68	0	0	1567	0	18	1
6	0	0	23	104	6	52	1907	29	102
7	0	6	108	2	18	476	20	452	181
8	0	1	26	34	5	5	106	0	3522

6.4 Lowest-Recall Classes

The three lowest-recall classes in the aggregate confusion matrix are:

- Class 7: recall 0.358
- Class 0: recall 0.724
- Class 6: recall 0.858

7 Discussion

The frozen `optimization` candidate achieved a high validation Macro-F1 mean (0.992459) during the train/validation phase, and the best validation Macro-F1 during tuning was 0.997589. However, the single one-time test evaluation produced a Macro-F1 mean of 0.834035, yielding a material validation-to-test gap of 0.158425. Seed-level analysis indicates that the gap is not uniform across seeds. Seeds 7 and 17 produced comparable test Macro-F1 values (0.875924 and 0.879813) and accuracies (0.906128 and 0.906267), both at best epoch 39. Seed 27, however, was unstable on test, achieving a Macro-F1 of only 0.746367 at best epoch 18, with a substantially higher loss (0.722006) compared to seeds 7 (0.397548) and 17 (0.412215). This instability reduces the mean and increases the variance of the aggregate test result. The aggregate confusion matrix indicates that class 7 has particularly low recall (0.358), with notable confusion into classes 5 (476 samples), 8 (181 samples), and 2 (108 samples). Class 0 recall is 0.724, with notable confusion into class 5 (683 samples) and class 1 (335 samples). Class 6 recall is 0.858, with notable confusion into class 3 (104 samples) and class 8 (102 samples). These observations are final reporting only and did not trigger further tuning, in accordance with the test policy that disallows tuning feedback and permits exactly one test evaluation.

8 Limitations

Several limitations follow directly from the supplied artifacts: 1. **Single test evaluation.** The test split was evaluated exactly once after candidate freezing. No repeated test evaluations were performed, and test results were not used for tuning, model selection, or retraining. 2. **Material validation-to-test gap.** The validation-to-test Macro-F1 gap was 0.158425, indicating a substantial transfer gap from validation to test. 3. **Seed instability.** Seed 27 was unstable on test, reducing the mean Macro-F1 and increasing the standard deviation. The small number of seeds (three) limits the robustness of the aggregate estimate. 4. **Class-level imbalance in errors.** Class 7 recall was 0.358, indicating severe under-performance on that class. Classes 0 and 6 also had reduced recall (0.724 and 0.858). 5. **No external benchmarking.** This report is constrained to the supplied frozen artifacts and does not compare against external models or

literature. 6. **No clinical or diagnostic claims.** This report makes no clinical use or diagnostic claims. The results are research reporting only.

9 Conclusion

We presented a frozen-candidate evaluation of a `SmallResNet optimization` variant on PathMNIST. The candidate was trained and tuned using only train/validation splits, and the test split was evaluated exactly once after candidate freezing. The best validation Macro-F1 during tuning was `0.997589`. The single one-time test evaluation produced a Macro-F1 mean of `0.834035` with standard deviation `0.075947`, and a validation-to-test gap of `0.158425`. Seed 27 was unstable on test, while seeds 7 and 17 produced comparable and higher Macro-F1 values. The three lowest-recall classes were 7 (`0.358`), 0 (`0.724`), and 6 (`0.858`). These observations are final reporting only and did not trigger further tuning. We do not propose clinical use or diagnostic claims.

10 Reproducibility Statement

10.1 Dataset

- Dataset: `PathMNIST`
- SHA-256: `1e7fc200dd5aac79f39f0da26178be801ffac8e6d59c8533e51acebae953745a`

10.2 Frozen Candidate

- Model: `SmallResNet`
- Variant: `optimization`
- Status: `approved_frozen`
- Approved by: `project_owner`
- Approved at: `2026-08-18`
- Seeds: `7, 17, 27`
- Learning rate: `0.001`
- Weight decay: `1e-05`
- OneCycle: `true`
- Label smoothing: `0.1`
- Augmentation: `false`
- Multiscale: `false`
- Primary metric: `macro_f1`

10.3 Test Policy

- Split: `test`
- Evaluations allowed: `1`
- Tuning feedback allowed: `false`
- Evaluation count: `1`

The test set was evaluated exactly once after candidate freezing. Test results were not used for tuning, model selection, or retraining.

10.4 Runtime Environment

- PyTorch: 2.13.0+cu132
- CUDA: 13.2
- GPU: NVIDIA GeForce RTX 5070 Ti Laptop GPU

10.5 Artifacts

- runs\pathmnist-m4\tuning\result.json
- runs\pathmnist-m4\final\optimization\seed_7\checkpoint.pt
- runs\pathmnist-m4\final\optimization\seed_17\checkpoint.pt
- runs\pathmnist-m4\final\optimization\seed_27\checkpoint.pt
- runs\pathmnist-m4\test_evaluation.json